WildChrokie WDC

**This is the WDC for the common garden project on tree cookies which will be included in Christophe's Chapter 2 **

Last Modified: 26 October 2026

Date of Study

2015 to spring 2023. Followed work on the cookies started in September 2024 and was submitted to New Phytologist in November 2026. Should be completed some time after this

Date of Public Archiving:

January 2027 (estimated)

Goal

Understand how longer and warmer season relates to growth for juvenile trees across different provenances.

Contributors

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- Victor Van der Meersch
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- Dan Buonaiuto
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General Files

File	Where	What
guidelinescookies.md	Github	Notes on how to use the tool Adam created to add guidelines to cookies. Python code can be found here
notesCookieScaping.md	Github	All the notes taken while measuring tree rings. This includes measurements rounds, the cookies for which I was confident and the ones I am not.
chaptersDesign.pdf	Github	Pdf visualization made in Illustrator for both CoringtreeSpotters and Wildchrokie
CoreXCookies.md	Github	Notes on cookie and core scraping, back when I was trying to confirm if the cookie was harvested in the spring 2023 or fall 2023. We knew for a fact that all cores were harvested in spring 2023, so i was confirming the measurements for the cookies with cores.
visualCrossdating.md	Github	Some notes on the visual cross dating process.
Outline CSEE talk.md	Github	Outline for CSEE2026 talk
notesGSLVSGrowth.xlsx	Github	Personal and non-exhaustive notes on the main papers looking at the effect of GSL on growth. Greffon repo has a much more thorough review
modelNotes.tex	Github	Notes on models after discussions with Victor, early on in the project that are irrelevant now.
letterGCB.tex	Github	Cover letter submitted to GCB
letterNPH.tex	Github	Cover letter submitted to New Phytologist
msWildchrokie.Rnw	Github	main Wildchrokie ms file without the supplement
supplement.Rnw	Github	supplement figures and tables of Wildchrokie
manuscriptOutline.Rnw	Github	Wildchrokie outline
resultsOutlineReal.Rnw	Github	Wildchrokie results outline

Data

File	Where	What
Seedlings to Raised Beds 2016.edit.xlsx	Github	Seedlings from raised beds. This is data coped from the main repo
Folder: ringWidth	Github	all the different tree csvs including the cores and cookies. They are split between cookies, cookiesUnconfident (these were verified and used in the analyses), coresUnconfident (these were not used since we only used the data for the cookies; the cores were only for visual cross dating; except for 2 cores for which we didn't have cookies for: folder coresWithoutCookies), coresWithCookies (used for visual cross dating, not used in the analyses).
climateNA.csv	Github	small dataset, manually extracted from:https://climatena.ca/mapversion
terraclimate42.2960N71.1206W.csv	Github	Terra climate mainly to get PDSI for the Arnold arboretum coordinates
Folder: hoboMicroStations	Github	From main repo
Daily Data for Boston (Weld Hill), MA - [id=ma_weld nwon, lat=42.2953, lon=-71.1337]	Github	daily weldhell climate data from 2019 for which I used 2020 since the other weldhill dataset stopped in august 2020
weldhill.csv	Github	weldhill cliamate station that stopped running in august 2020. Used for GDD calculations for main analyses
bostonAirportClimate.csv	Github	Long term climate to calculate the gdd for the conceptual figure
treecookies.xlsx	Github	Old file that helped for scraping cookies

Code

File	Where	What
getAll.R	Github	sources all the four files below, compiles them to yield the main data file: `empiricalDataMAIN.csv` and `empiricalData_pub.csv` which is a shorter of the former for publication.
calculateGrowingSeasonGDD.R	Github	**Needs to be run through getAll.R** this code uses the climate data cleaned and binded in `cleanClimate.R` and with the pollen package, it calculates the growing degree days first, then calculates the GDD between leafout and budset (for the primary season) or between budburst and leaf fall (for the full growing season---though this metric isn't used in the analyses)
cleanClimate.r	Github	Checks old climate station at weld hill that stopped running in august 2020 and new one that started in January and cleans the data, combining both dataset to have the years we're interested in. Also uses terra climate to clean and get 3 month periods of PDSI and other summarized data metrics.
cleanObsData.R	Github	Cat and Dan's observational data cleaning code. Takes all the observations from 2018 to 2020, cleans it to yield a doy per individual per year, with columns for each phenophase
cleanRingWidth.R	Github	

Folder: fromMainRepo	Github	This is a dupplicate from wildhellgarden repo, so the info for that folder is in that WDC.
growthModelsMain.R	Github	This is the main code for the analyses. All four models are calculated in this code. It includes the model runs, posterior predictive checks, and prior vs posterior checks and other diagnostics. It includes also all the "check" models such as the full vs restricted data model comparison.
growthModelsMain_plots.R	Github	Sources `growthModelsMain.R` and makes all the figures used in the manuscript (and many more) except for the conceptual figure which is made in `GSfigure.R`
GSfigure.R	Github	Makes the conceptual figure used in Wildchrokie MS
plottingEmpiricalData.R	Github	Non-modelled data visualization. Most of it was for initial data visualization, but it also have some allometry plots that we included in the MS
climate_plots.R	Github	Some climate plots that we don't use anymore, but some of that code could still be useful to calculate frost free days, PDSI, for instance.
tools.R	Github	Repetive code that I removed that I merged into functions with Claude to make the codes more efficient. This includes, color palets and posterior recovery functions.
modelGrowthPreviousYear.R	Github	Outdated as we decided to drop this model. It was using the previous year GDD as another predictor. We kept the

		idea in CoringTreespotters however.
powerPointPlots.R	Github	Plots broken down for my presentation at CSEE 2026
modelGrowthYear.R	Github	Simple growth model as a function of year. We wanted to see if a year only model was more predictive than our models with other predictors. We used loo for this.
growthModelsMainbaseline.R	Github	Side quest to determine wether we wanted baselines in the models. We decided to drop to make the analyses more interpretable by a broad public
carryOverLeafoutBudset.R	Github	Side quest to determine if there was a carry over effect of leafout on budset. Bad PPC, so we dropped the idea as we weren't that interested in it in the end.
checkGenQuantities.R	Github	Old code to check if my generated quantities worked fine.
proveNestedEquivalence.R	Github	We've been wondering wether the nested model and non nested model were doing the same thing. See : https://github.com/christophe-rd/wildchrokie/issues/18 for discussion on the subject.
modelGrowthSOS.stan	Github	One of the four main models, using start of season (leafout) as the growing season predictor.
modelGrowthEOS.stan	Github	One of the four main models, using end of season (budset) as the growing season predictor.
modelGrowthGSL.stan	Github	One of the four main models, using growing season length (number of days between leafout and budset) as

		the growing season predictor.
modelGrowthGDD.stan	Github	One of the four main models, using growing degree days (thermal accumulation above 5C between leafout and budset) as the growing season predictor.
modelGrowthGDD_asitePerSpp.stan	Github	One of the model check where we fit provenance parameter for each species
modelGrowthzlargerPriors.stan	Github	Larger prior checks with z-scored predictors. Didn't use in paper
modelGrowthGDD_bMax30.stan	Github	Side quest with a slope on temperatures above 30C. Didn't use in paper
modelGrowthGDD_BAI.stan	Github	GDD model using BAI as a response variable
modelGrowthGSL_BAI.stan	Github	GSL model using BAI as a response variable
modelGrowthSOS_BAI.stan	Github	SOS model using BAI as a response variable
modelGrowthEOS_BAI.stan	Github	EOS model using BAI as a response variable
modelGrowthPreviousYear.stan	Github	Previous year model that we did in TS, but didn't work here.
modelGrowthGDD_PPsite.stan	Github	Check before we decided to move on with partial pooling on asite for good
modelGrowthGDD_ayear.stan	Github	Check before we decided to move on with an intercept on year for good.